

GIRIDHARAN ANBARASAN

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[Portfolio](#)

Location: New Jersey, USA

SUMMARY

Software Engineer with 7+ years of experience building scalable backend systems, data platforms and AI-powered solutions. Proficient in Python, Java and Go, with expertise in REST APIs, microservices, pipeline automation and data engineering. Experienced in machine learning, predictive modeling, LLMs, building intelligent automation and data-driven applications. Proven track record of improving system performance, automating workflows, and building scalable platforms.

PROFESSIONAL EXPERIENCE

SENIOR SOFTWARE ENGINEER - ORACLE, USA

Jul 2024 – Apr 2026

Skills: Terraform, Go, Python, MySQL, LangChain, LLMs, Generative AI, REST APIs, Pipeline Automation, CI/CD, OCI, Git, Bitbucket, Bash

- Improved resource discovery performance by 3X by leveraging Go routines and optimized concurrency patterns, accelerating Terraform onboarding.
- Built and deployed an AI Slack assistant using Python and LangChain that routes queries, retrieves context from internal knowledge bases, and resolves support requests for 100+ teams, reducing turnaround time by 40%. Stored interaction logs in MySQL to identify knowledge gaps and improve the knowledge base.
- Built and deployed a Python-based pipeline monitoring and alerting system, storing logs in MySQL for failure analysis, reducing mean time to detection by 30%.
- Built and delivered a unified Terraform provider, combining 100+ internal service providers into one, reducing build time by 80% and deployment time by 70%.
- Designed and implemented automated PR review workflows across Terraform Provider repositories, reducing review turnaround time by 2 weeks per team.
- Developed and enhanced developer tooling for setup, development, and testing of Terraform providers, reducing onboarding time by 1 week per team.

DATA SCIENTIST, ICREDITWORKS, USA

Jan 2022 – Jun 2024

Skills: Python, SQL, Snowflake, PySpark, Apache Airflow, REST APIs, FastAPI, ETL, Data Engineering, Data Modeling, Docker, Power BI, Automation, Git

- Led the migration from OLTP-based analytics to a Snowflake OLAP platform. Built Python-based Airflow pipelines to ingest data into Snowflake and used Snowpark (PySpark) to create dimension tables, fact tables, and analytical views, enabling scalable reporting.
- Built end-to-end backend systems for web analytics using PySpark for large-scale data processing, Airflow for data orchestration, and FastAPI to serve precomputed results, reducing Power BI licensing costs by up to \$150K annually.
- Implemented time-series forecasting models using Facebook Prophet to predict KPIs, enabling proactive planning and data-driven decision-making.
- Redesigned and automated analytics data pipelines with Apache Airflow, and migrated reports to the new ecosystem.
- Created interactive Power BI dashboards to visualize KPIs, enabling data-driven decision-making by translating complex reports into actionable insights, used by business stakeholders for operational and strategic decision-making.
- Automated daily financial activity and summary reports using Python, with email delivery via SMTP and SFTP to improve reporting efficiency and accuracy.

DATA SCIENTIST INTERN - MILLIPORESIGMA, USA

Jun 2021 – Dec 2021

Skills: Python, Machine Learning, Data Engineering, ETL, Predictive Modeling, Data Analysis, Palantir Foundry

- Improved on-time delivery rate from 81% to 95% using machine learning based predictive models, delivering €1.18M in annual efficiency gains.
- Analyzed over 1M records of historical biotesting data to assess time spent across each phase of the testing lifecycle.
- Developed a hierarchical machine learning model to predict study durations at each stage, improving workflow planning and resource allocation.
- Deployed the ML model within a production application to monitor status of studies, identify at-risk projects, and prioritize them to ensure on-time delivery.

SOFTWARE ENGINEER - ZOHOO CORPORATION, INDIA

Jan 2017 – Dec 2019

Skills: Java, SQL, Spring Boot, REST APIs, Kafka, Redis, Microservices, Event-Driven Architecture, Real-time updates, JUnit, JavaScript, HTML, CSS, Docker

- Implemented two-way integration between Zoho Desk and Zoho Chat using Kafka, enabling real-time customer interactions via chat widgets and chatbots, reducing support queries by ~70% after release.
- Designed and developed a gamification service to encourage agents by awarding points and badges based on response time, resolution speed, and escalations, improving customer handling and service quality.
- Developed backend services for Manager Dashboard using Redis to maintain real-time customer state, with updates driven by web messaging systems, enabling low-latency monitoring of online status, ticket counts and customer satisfaction.
- Developed an Agent Collision prevention system using Redis to track real-time ticket activity and alert agents when multiple users interacted with the same ticket, reducing duplicate work and improving resolution efficiency.

PROJECTS

PERSONAL FINANCIAL ANALYTICS PLATFORM

- Developed a private financial analytics platform to ingest and analyze transaction data, providing insights into spending, recurring expenses, and cash flow.
- Developed rule-based transaction categorization to group expenses and implemented Isolation Forest to detect unusual transactions and spending patterns.
- Developed an LLM-powered analytics layer for natural-language queries across transactions, subscriptions, merchants, categories, and spending trends.
- Exposed core functionality through FastAPI and PostgreSQL, supporting transaction ingestion, analytics, search, and user interactions.

SMART HOUSEHOLD MANAGEMENT SYSTEM

- Developed a smart assistant that turns household documents such as insurance policies, warranties, bills, registrations, and receipts into organized records.
- Designed an OCR and LLM pipeline to identify document types and extract details such as coverage, amounts, expiration dates, renewal terms, and warranties.
- Implemented natural-language search across both document content and structured records, with RAG to provide answers grounded in the user's documents.
- Created a lifecycle tracking system that turns dates and terms from documents into reminders for renewals, expirations, warranties, and household maintenance.
- Exposed the core functionality through REST APIs and PostgreSQL, supporting document ingestion, extracted information, search, tasks, and user interactions.

SKILLS

- PROGRAMMING & BACKEND FRAMEWORKS** - Python, Java, Go, SQL, R, JavaScript, HTML, CSS, Spring Boot, REST APIs, FastAPI, Sanic, JUnit, MySQL, PostgreSQL.
- AI & ML** - LLMs, Gen AI, LangChain, RAG, OCR, Isolation Forest, Time-Series Analysis, Pandas, Scikit-learn, PySpark, TensorFlow, Image Processing
- CLOUD & INFRASTRUCTURE** - AWS, GCP, OCI, Snowflake, Docker, Kubernetes, Terraform, Apache Airflow, Kafka, Redis
- TOOLS & PLATFORMS** - Git, Bitbucket, Jira, Confluence, Postman, MySQL Workbench, Palantir Foundry, Power BI, Tableau.

EDUCATION

MASTERS IN DATA SCIENCE - NEW JERSEY INSTITUTE OF TECHNOLOGY

Jan 2020 - Dec 2021

BACHELORS IN INFORMATION TECHNOLOGY - ANNA UNIVERSITY, INDIA

Aug 2013 - May 2017

CERTIFICATION

Google TensorFlow Developer Certificate

AWS Machine Learning Specialty Certificate

Meta Front End Developer Certificate